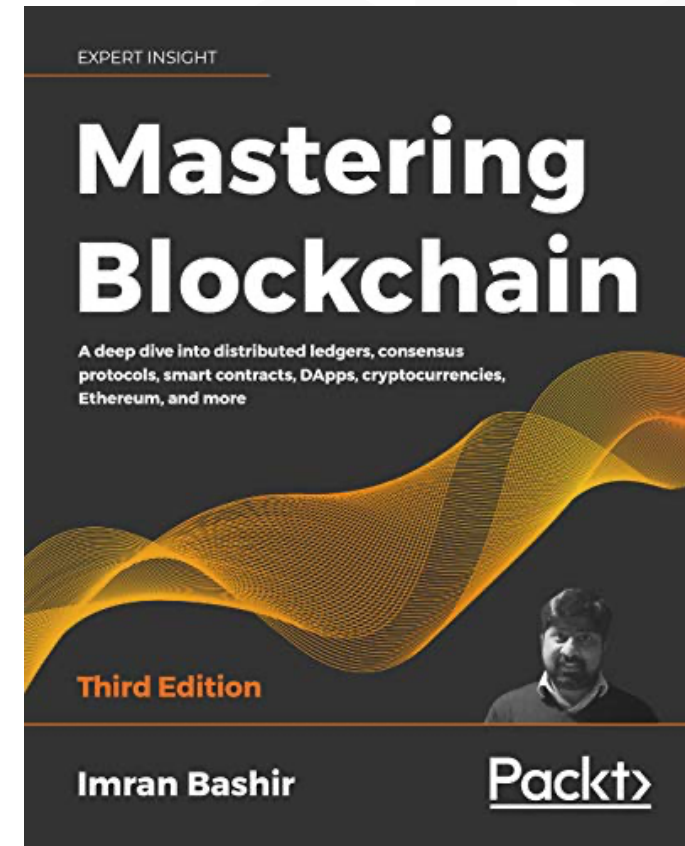


Mastering Blockchain

Third Edition

Chapter 7, Bitcoin Network and Payments



Outline



- The Bitcoin network
- Clients
- Discovery
- Wallets
- Bitcoin payments
- Innovation in Bitcoin

Nodes



- Node types
 - Full nodes
 - SPV (Simple Payment Verification)
- Pool protocols
 - Stratum

Bitcoin network



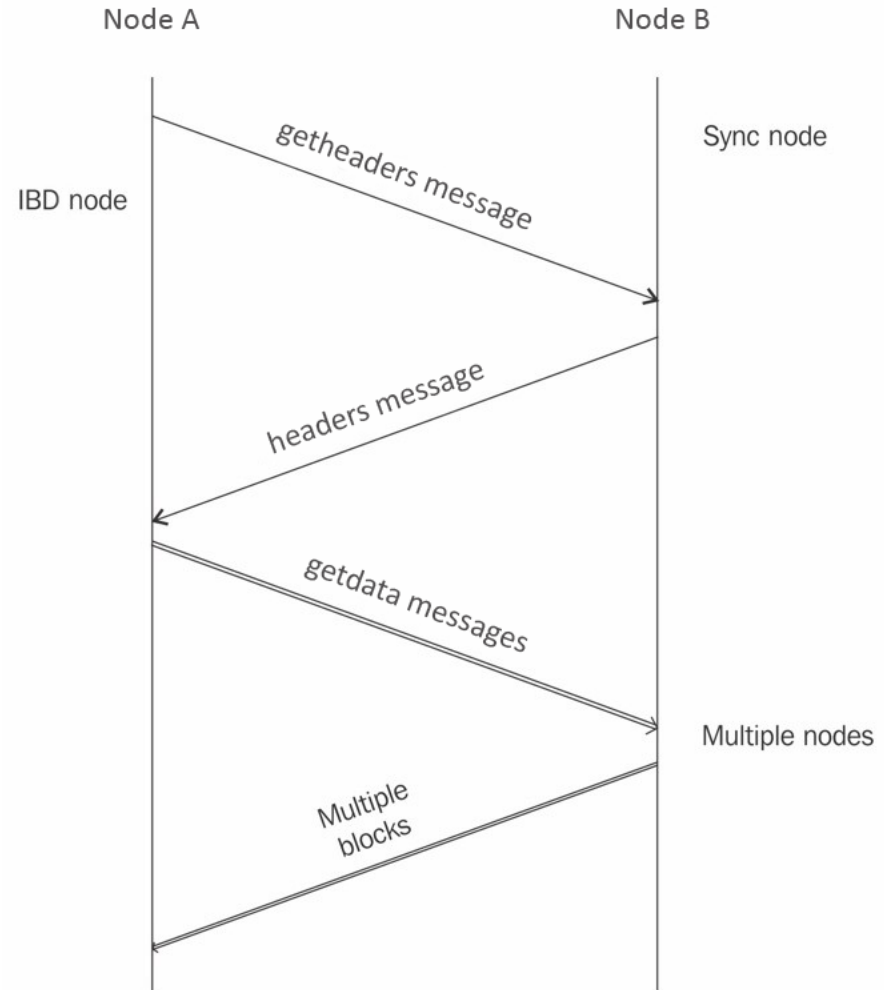
A Bitcoin network is identified by its magic value. Magic values are used to indicate the message's origin network.

Network	Magic value (in hex)
main	0xD9B4BEF9
testnet	0xDAB5BFFA
testnet3	0x0709110B

Protocol messages

- Version
- Verack
- Inv
- Getdata
- Getblocks
- Getheaders
- Tx
- Block
- Headers
- Getaddr
- Addr
- Ping
- Pong

This shows the Bitcoin block synchronization process between two nodes on the Bitcoin network.



Bloom filters



A Bloom filter is a data structure which is used to find out whether an element is present in a set.

- Bloom filters are quick and memory efficient. Due to efficiency requirements they are probabilistic in nature.
- Bloom filters provide probabilistic lookup with false positives but no false negatives.
- `fnv`, `murmur`, and `Jenkins` are commonly used hash functions in bloom filters due to their speed and uniformity.
- Used in SPV Bitcoin clients.
- Bitcoin Improvement Proposal 37, implemented in Bitcoin as BIP37, introduced three new message types : `filterload`, `filteradd` and `filterclear`.

Wallets



- Non-deterministic wallets
- Deterministic wallets
- Hierarchical deterministic wallets
- Brain wallets
- Paper wallets
- Hardware wallets
- Online wallets
- Mobile wallets

Bitcoin payments

Bitcoin can be accepted as payment using various techniques.
Bitcoin is not recognized as a legal currency in many jurisdictions, but it is increasingly being accepted as a payment method by many online merchants and e-commerce websites.



Bitcoin payment sign, usually used at stores where Bitcoin payments are accepted



Point of sale terminal

Innovation in Bitcoin



- Bitcoin Improvement Proposals (BIPs)
 - Standard BIP
 - Process BIP
 - Informational BIP
- Advanced protocols
 - SegWit
 - Bitcoin Cash
 - Bitcoin Unlimited
 - Bitcoin Gold

Exercise



- Consider how a Bitcoin ATM could be designed.

Summary



In this presentation, we explored:

- The Bitcoin network
- Bitcoin node discovery
- Block synchronization protocols
- Network messages
- Bitcoin wallets
- Bitcoin payments and payment processors
- BIPs
- Advanced Bitcoin protocols